EX.NO.10
PRECISION SHOTS: LEVERAGING MACHINE LEARNING FOR SNIPER BALLISTICS OPTIMIZATION

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OUTLINE OF PRESENTATION

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INTRODUCTION

- The accuracy and precision of sniper shots can mean the difference between success and failure.
- Snipers rely on advanced ballistic calculations to compensate for factors like windage, bullet drop, and environmental conditions when engaging targets at long distances.
- Traditional methods of calculating ballistic trajectories often require manual adjustments and are subject to errors, highlighting the need for more efficient and accurate solutions.

CONCEPT

The Machine Learning Arsenal:

"Precision Shots" are a myriad of ML algorithms strategically chosen to tackle the multifaceted challenges of sniper ballistics optimization:

Regression Algorithms:

- Linear regression serves for capturing linear relationships between input variables like muzzle velocity, wind speed, and output variables.
- Polynomial regression adds a layer of complexity, while support vector regression (SVR) adapts to nonlinearities, providing a comprehensive spectrum of predictive capabilities.

PROBLEM STATEMENT

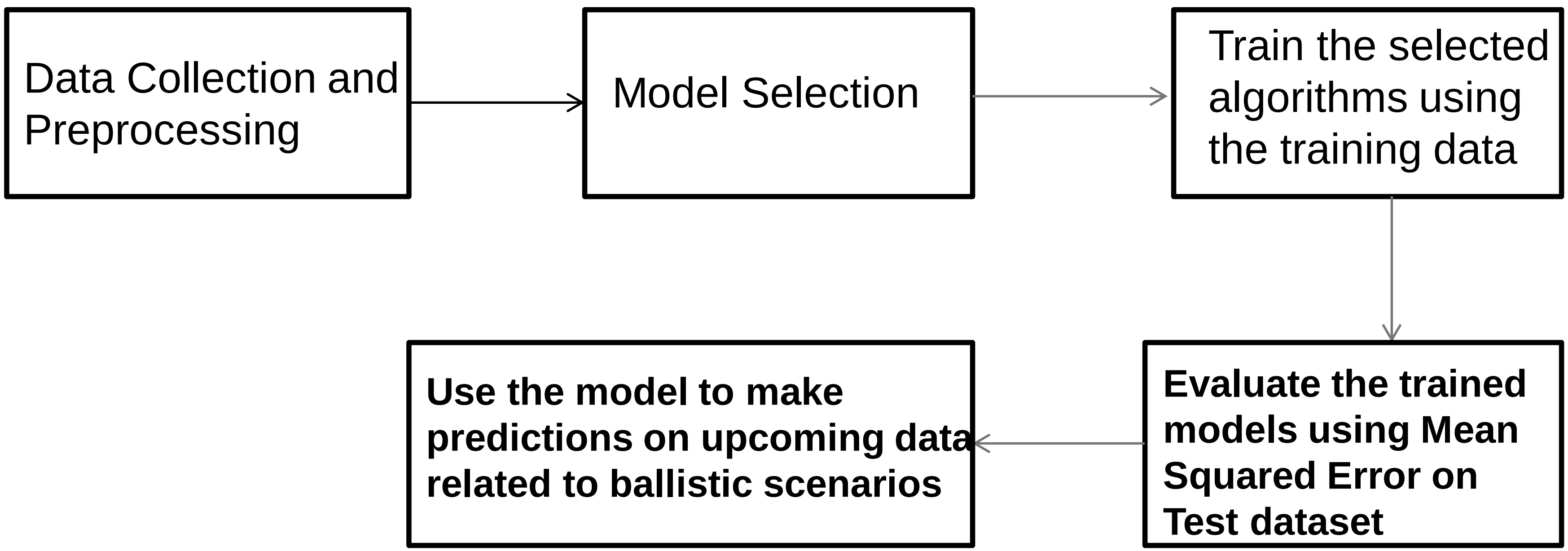
Problem Statement: Develop a machine learning model to accurately predict ballistic trajectories and optimize shooting solutions for snipers in various environmental conditions.

- ML algorithms can learn to predict the trajectory of a bullet based on factors like muzzle velocity, bullet weight, wind speed/direction and target distance.
- This information is crucial for snipers to make accurate shots over long distances.
- ML algorithms can analyze wind data and learn to adjust for windage, helping snipers compensate for crosswinds that can deflect the bullet's path

KEY OBJECTIVES:

- Collect and preprocess a dataset containing relevant ballistic data, including muzzle velocity, wind speed, and corresponding trajectory/impact points.
- Explore and analyze the dataset to understand the relationships and dependencies between input and output variables.
- Implement and evaluate multiple machine learning algorithms, including linear regression, polynomial regression, SVR, and random forests, to determine their effectiveness in predicting bullet trajectories and impact points.
- Validate the predictive model using appropriate evaluation metrics and testing datasets to ensure its accuracy and generalizability across different scenarios.

FLOW DIAGRAM OF THE MINI PROJECT



METHODOLOGY

- Linear Regression: Models linear relationships between input features and output (trajectory/impact points).
- Polynomial Regression: Captures non-linear relationships by introducing polynomial terms.
- SVR : Handles complex relationships and non-linear data patterns.
- Random Forest: Ensemble method for predictions, especially useful for high-dimensional data.
- Reason for Choosing: Each algorithm offers unique strengths for capturing different types of relationships and correlation while handling various data complexities.

DATA SET

The Dataset used for this project where collected from multiple sources and tabled

https://bergerbullets.com/nobsbc/muzzle-velocity-vs-ballistic-coefficient-trade-offs/

https://www.backcountrychronicles.com/ballistic-factors-bullet-wind-drift/

it is also a major limitation of this project, as there isn't a lot of publically available dataset that correlates windspeed, velocity and .

	A	B	C	D
1	muzzle_v	wind_spe	bullet_impact	
2	2600	0	-2	
3	2398.1	0.4	0	
4	2205.1	1.7	-4.1	
5	2205.5	4.1	-15.3	
6	2022.3	7.6	-35	
7	1847.5	12.4	-65	
8	1680.1	18.8	-107.3	
9	1519.5	27	-164.8	
10	1366	37.3	-240.9	
11	1221.3	50	-340.5	
12	1093.2	64.8	-469	
13	1029.8	55.6	-474.2	
14	1022.5	63.6	-480.6	
15	1022	65.4	-345.5	
16	1033	67.7	-320.6	
17	1031.4	64.5	-314.2	
18	1024	14.4	-97.2	
19	1022.4	20.2	-108.5	
20	1035.7	25	-133.7	
21	1077.3	17.4	-102.5	
22	1043.4	19	-107.7	
23	1075.8	18.5	-106.7	
24	1078.9	14.4	-97.8	
25	1055.2	9.6	-46	

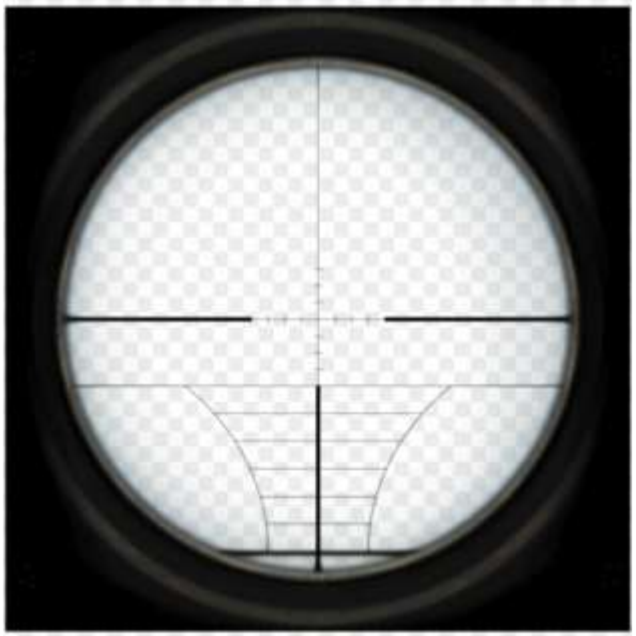
ballistic_data

RESULTS AND DISCUSSION

- To the part that actually matters. I've attached an image of what an output would look like

```
Choose Files ballistic_data.csv
• ballistic_data.csv(text/csv) - 471 bytes, last modified: 5/15/2024 - 100% done
Saving ballistic_data.csv to ballistic_data (1).csv
Linear Regression MSE: 4166.3170212225405
Polynomial Regression MSE: 5192.684492050379
SVR MSE: 31640.370330731977
Random Forest MSE: 2849.1028976000057
```

- It might look like just numbers and decimals but it holds sometime more. Let me explain



- Now using our regression model, we predict where the bullet is heading with x input variables such as mussle_velocity and wind_speed(with direction represented as positive or negative based on the horizon)
- And we also have y input variable bullet_impact(representing where the bullet ended up)
- Now with this combined knowledge we use a MSE function to find the difference between our predicted value and the actual value.

CONCLUSION

- The core design of "Precision Shots" encapsulates a holistic approach to sniper ballistics optimization, blending cutting-edge machine learning techniques with real-world operational requirements.
- "Precision Shots" does not merely aim at algorithmic prowess; it aspires to redefine the landscape of sniper operations.
- By arming snipers with data-driven insights, real-time predictions, and adaptive strategies, the project envisions heightened mission success rates, reduced manual calculation efforts,.
- a transformative leap towards precision, efficiency, and tactical superiority on the battlefield.

BIBLIOGRAPHY

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1. https://gundata.org/ballistic-calculator

WEBSITES

1. https://fusilero.github.io/balistica/
2. https://pypi.org/project/py-ballisticcalc
3. https://shooterscalculator.com/ballistic-trajectory

Questions?

Thank you